ICT303 – Advanced Machine Learning and Artificial Intelligence

Topic 5: Convolutional Neural Networks (CNN) – Part I

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How to Get in Touch with the Teaching Team

- Internal and External Students
  - Email: H.Laga@murdoch.edu.au.
- Important
  - In any communication, please make sure that you
    - Start the subject of your email with ICT303
    - Include your student ID, name, and the lab slot in which you are enrolled.
  - We will do all our best to answer your queries within 24 hrs.

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In this Lecture

- Introduction
  - Recap of Linear regression, perceptron and MLP
  - Why we need CNNs?
- Convolutional Neural Network (CNN)
  - Convolutions
  - Convolution layers
- Examples of CNNs
  - LeNet
- Summary

- Learning objectives
  - Understand CNN and the different components that compose it
  - Implement CNNs in Python, NumPy and PyTorch
- Additional readings
  - Chapter 7 of the textbook, available at: https://d2l.ai/

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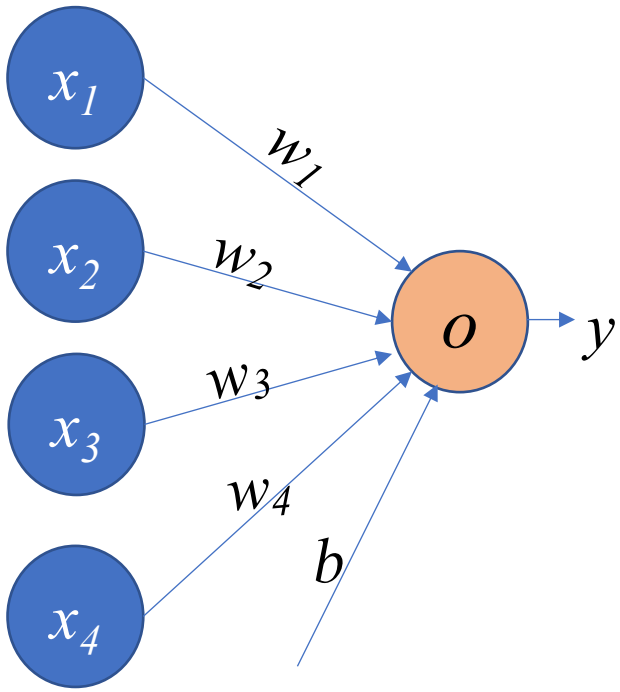
Linear Methods for Regression and Classification (Recap)

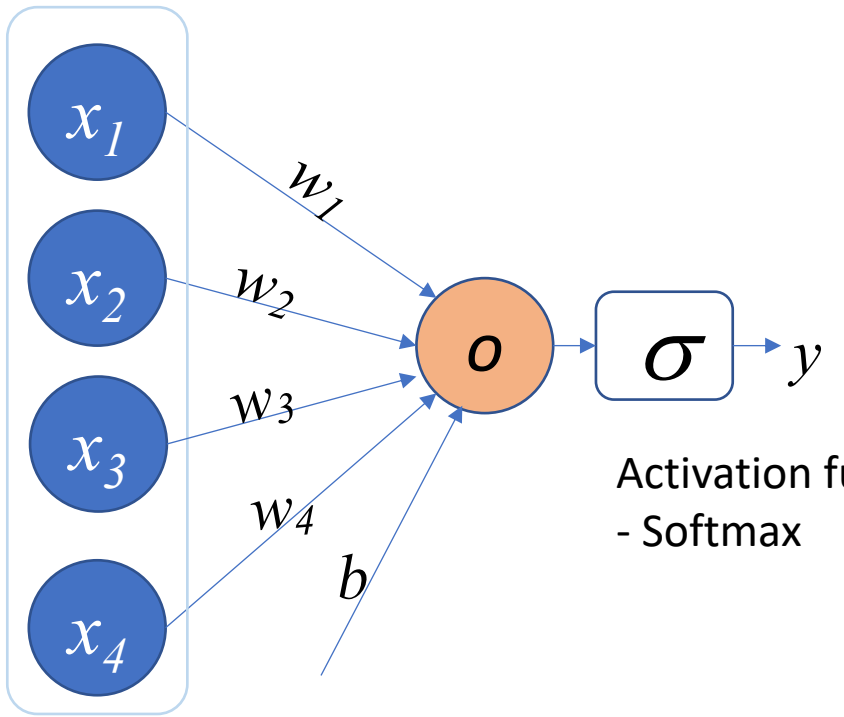
- Linear regression

A dense (fully connected, or linear) layer has parameters $\mathbf{W} \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$, it computes output $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b} \in \mathbb{R}^m$



$$o = y = \langle W, X \rangle + b$$

- Classification



Activation functions
- Softmax

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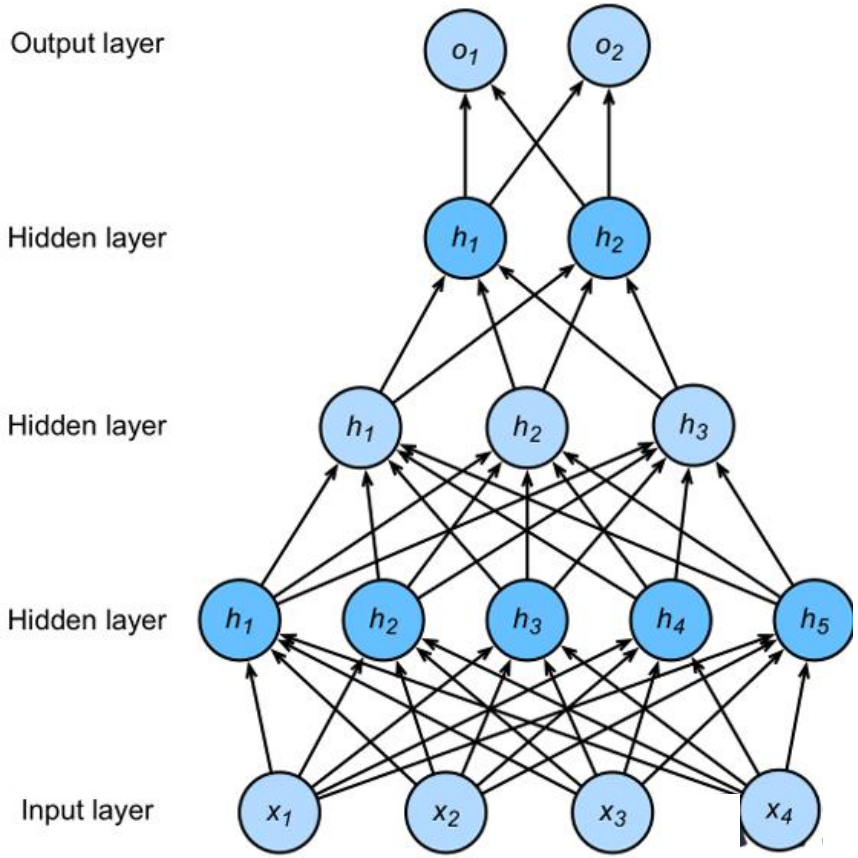
Multilayer Perceptron (MLP)

- Stacks multiple hidden layers (dense + activation) to get deeper models
- The activation function leads to non-linear models
  - Sigmoid, ReLU
- Hyper parameters
  - No. of hidden layers
  - No. of outputs of each hidden layers
- They are universal approximators
  - Can approximate any function



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MLPs limitations

- Design an MLP composed of one single hidden layer
  - Input: RGB image of size 300 x 300 pixels
  - Output: The class of the object depicted in the image
  - Consider the case where we:
    - have 1000 possible object classes (horse, dogs, tables, ….)
    - use a simple MLP composed of 1 input layer, two hidden layers, and one output layer

- Questions
  - How many parameters the network will have?
  - If we randomly shuffle the image pixels, what would be the output produced by the network?

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MLPs limitations

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For 1 Single Hidden Layer (for 1 training example):

x = (300 x 300 x 3,1)
W = (1000,300x300x3)
b = (1000,1)
z = Wx + b

Total Parameters = 1000 x 300 x 300 x 3 + 1000
= 270 001 000

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From MLPS to Convolution Neural Networks (CNN)

- Issues with MLPs
  - Images are structured into grids of pixels
    - MLPs treat them as a flattened vector ignoring the local relations between pixels
    - If you take an image and randomly shuffle its pixels, the MLP will produce the same result!
  - Have a very large number of parameters
    - Thus, they are difficult to train for complex problems

- Convolutional Neural Networks (CNN or ConvNets)
  - A powerful family of neural networks specifically designed to
    - Capture the local spatial relationships between pixels
    - Reduce the number of parameters compared to MLPs
  - Uses convolutions

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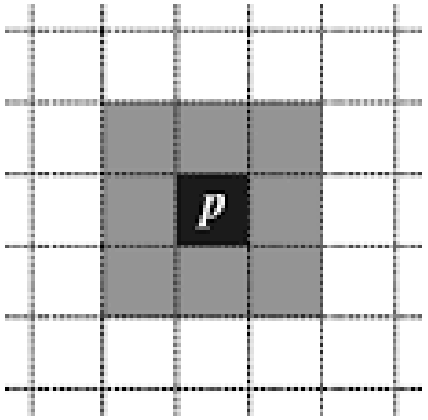
Convolution



A pixel p and its
8 neighbors

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

A 3 x 3 filter

$$\begin{aligned} J(x,y) = \; & aI(x-1,y-1) + bI(x,y-1) + cI(x+1,y-1) + \\ & dI(x-1,y) + eI(x,y) + fI(x+1,y) + \\ & gI(x-1,y+1) + hI(x,y+1) + iI(x+1,y+1) \end{aligned}$$

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Convolution - Example

Pixel Values

1	0	4	2	125	67
8	2	5	4	34	12
20	13	25	15	240	2
76	8	6	6	100	76
34	66	134	223	201	3
255	123	89	55	32	2

Kernel 3 x 3 Pixels

1	2	1
2	4	2
1	2	1

2 x 1 = 2
2 x 5 = 10
1 x 4 = 4
2 x 13 = 26
25 x 4 = 100
15 x 2 = 30
8 x 1 = 8
6 x 2 = 12
6 x 1 = 6

Convoluted Image

		198			

2 + 10 + 4 + 26 + 100 + 30 + 8 + 12 + 6 = 198

Image courtesy of: https://dev.to/sandeepbalachandran/machine-learning-convolution-with-color-images-2p41

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Illustration of Convolution

Given this 5x5 image, convolve with a 3x3 filter

1	2	2	3	1
1	1	3	3	2
3	2	1	2	3
2	3	2	3	2
2	1	2	3	1

1	2	1
1	2	1
1	2	1

[OFFICIAL]

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1	2	1	3	1
1	2	1	3	2
1	2	1	2	3
2	3	2	3	2
2	1	2	3	1

Move the filter over the entire image from left to right, top to bottom

1x1 + 2x2 + 1x2 + 1x1 + 2x1 + 1x3 + 1x3 + 2x2 + 1x1

21		

1	2	2	3	1
1	1	3	3	2
3	2	1	2	3
2	3	2	3	2
2	1	2	3	1

[OFFICIAL]

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Given this 5x5 image, convolve with a 3x3 filter

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2	1	2	3	1

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1x2 + 2x2 + 1x3 + 1x1 + 2x3 + 1x3 + 1x2 + 2x1 + 1x2

21	25	

1	2	2	3	1
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21	25	27

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21	25	27
25		

[OFFICIAL]

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21	25	27
25	26	

1	2	2	3	1
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[OFFICIAL]

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[OFFICIAL]

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21	25	27
25	26	29
24	24	27

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Convolution - Example

- Edge detection with Sobel filter (or Sobel kernel)
  - Convolving the image I with this operator will result in Jv, the image of vertical edges

-1	0	1
-2	0	2
-1	0	1

  - Convolving the image I with this operator will result in Jh, the image of horizontal edges

1	2	1
0	0	0
-1	-2	-1

- Combine the two filters to give a single measure of gradient magnitude

$$J(x,y) = \sqrt{J_h(x,y)^2 + J_v(x,y)^2}$$

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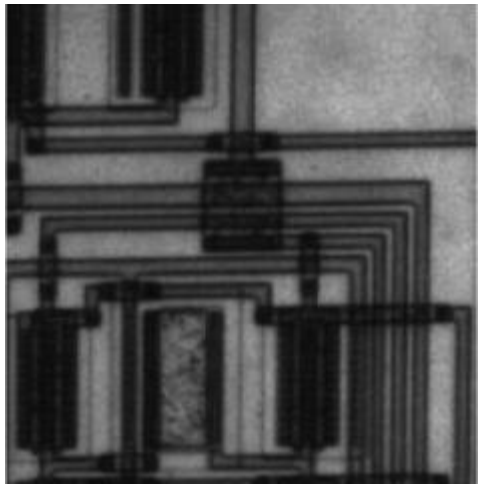
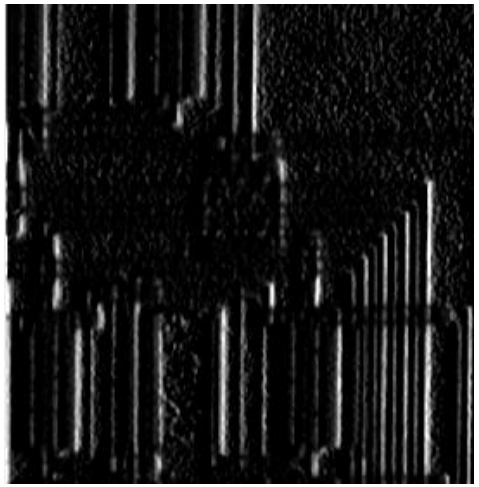
Convolution - Example

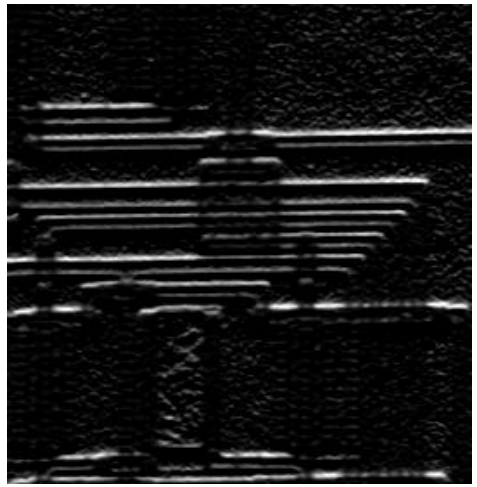
- Edge detection with Sobel filter (or Sobel kernel)



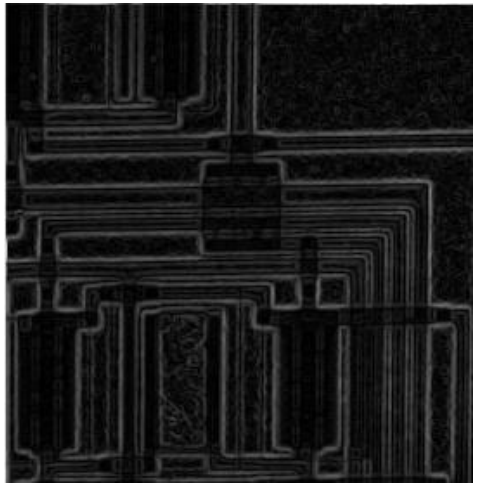
Input image



Vertical edges Iy



Horizontal edges Ix



$$G = \sqrt{Ix^2 + Iy^2}$$

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Convolution

- Question
  - Can you implement the Edge detector using neurons?

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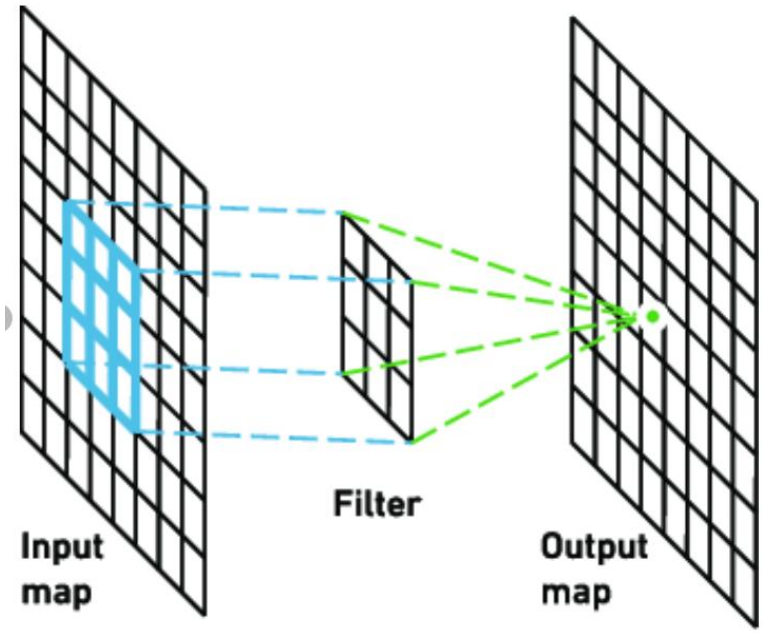
Convolution Layer

- At each image pixel p, unlike dense, or fully-connected, layer, convolutional layers take a weighted sum of the pixels around p
  - Think about it as a filter BUT the values of the filter (kernel) are unknown
  - The goal of training is to learn the values of the elements of the filter that best suit the task at hand



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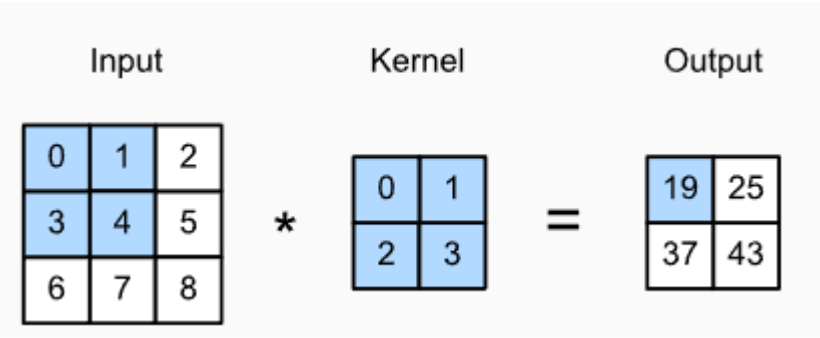
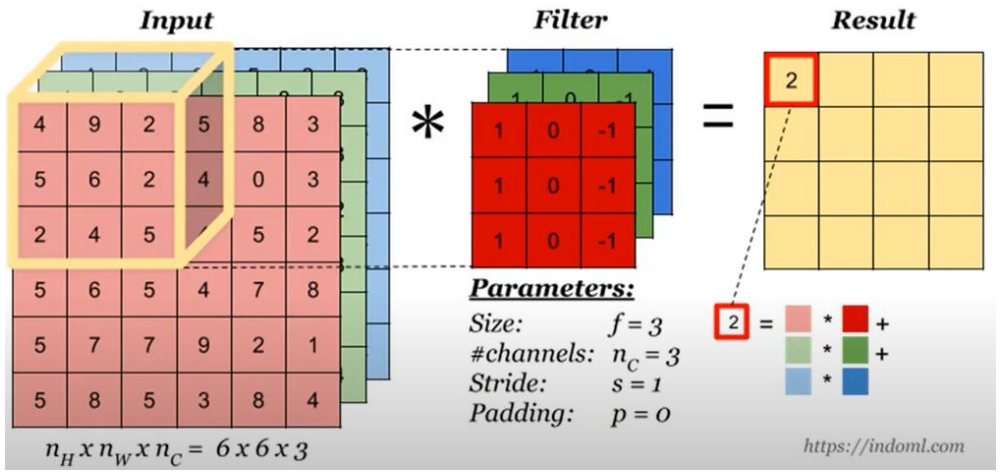
Convolution Layer – one filter case

- Greyscale image
  - Has d = 1 channel (for the grey level)
  - The filter is then of size s x s (e.g., s can be 3, 5, 7, ..)



https://d2l.ai/chapter_convolutional-neural-networks/conv-layer.html

- Color image
  - Has d=3 channels (one for R, one for G, and another for B)
  - The filter is then of size d x s x s (e.g., s can be 3, 5, 7, ..)



https://www.youtube.com/watch?v=3myNsOGhc3A&list=PLbNJQ-D5RH18U7BXm3NH8_6mnguQ45uRh&index=5

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Convolution Layer – one filter case

- Convolutional layer
  - Cross correlates the input image and the kernel and adds a bias to produce an output
- The two (learnable) parameters of a convolutional layer are
  - The kernel values
  - The bias
- The hyper parameters (not learnable, set by the user) are
  - The kernel size
    - It defines the receptive field size
    - Receptive field refers to all the elements from the previous layer that may affect the calculation of the output value
- The output of a convolutional layer is called feature map

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Convolution Layer – Multiple Feature Maps

- Apply different kernels to the input
  - Each kernel will produce one output, called feature map
  - Stack the feature maps together to produce a multichannel output

- Hyper parameters
  - The kernel size
    - It defines the receptive field size
    - Receptive field refers to all the elements from the previous layer that may affect the calculation of the output value
  - No. of output channels at each layer
    - It defines the no. of different filters to learn at each layer

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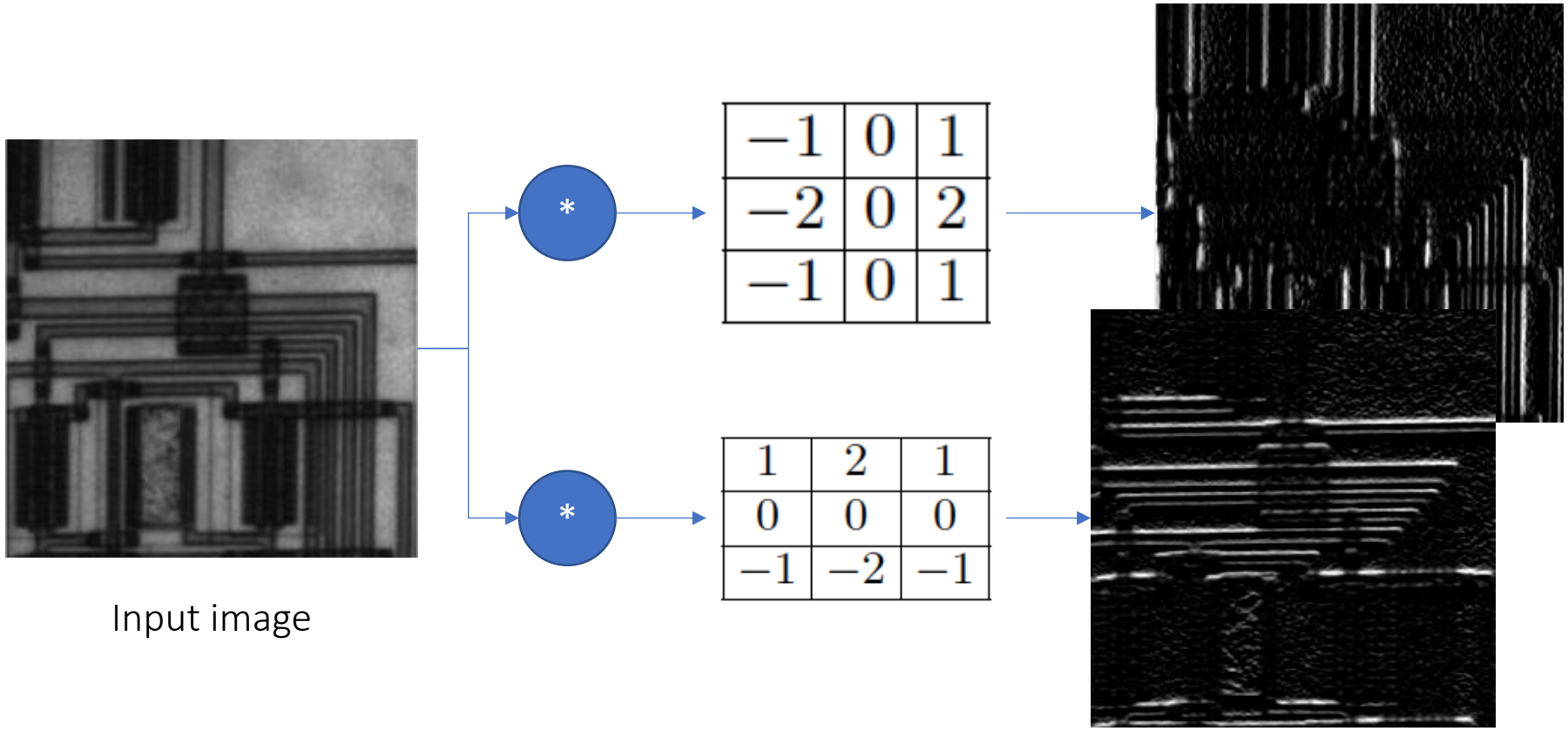
Convolution Layer – Multiple Feature Maps

- Example – Edge detection with Sobel filter (or Sobel kernel)



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Convolution Layer – Practical Considerations

- Padding
  - When you convolve an image with a filter, we will lose pixels at the boundary of the image

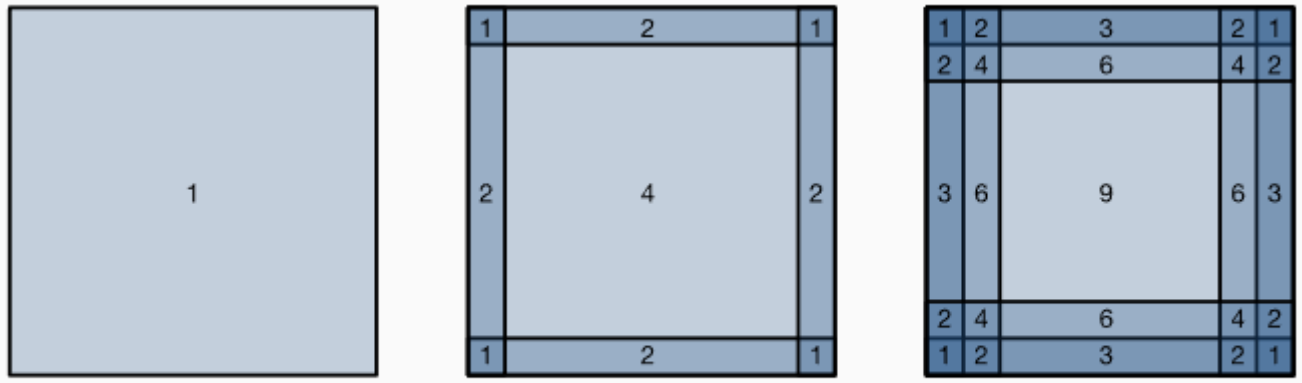

Fig. 7.3.1 Pixel utilization for convolutions of size 1×1, 2×2, and 3×3 respectively.

https://d2l.ai/chapter_convolutional-neural-networks/padding-and-strides.html

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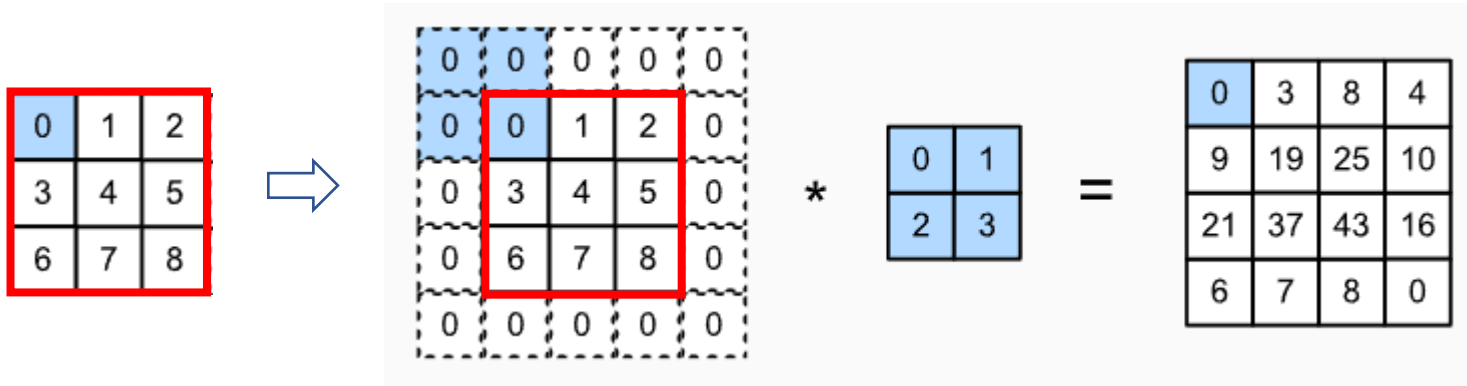
Convolution Layer – Practical Considerations

- Padding
  - When you convolve an image with a filter, we will loose pixels at the boundary of the image
- Solution → Padding
  - Add extra pixels (set to 0) of filler around the boundary of the input image



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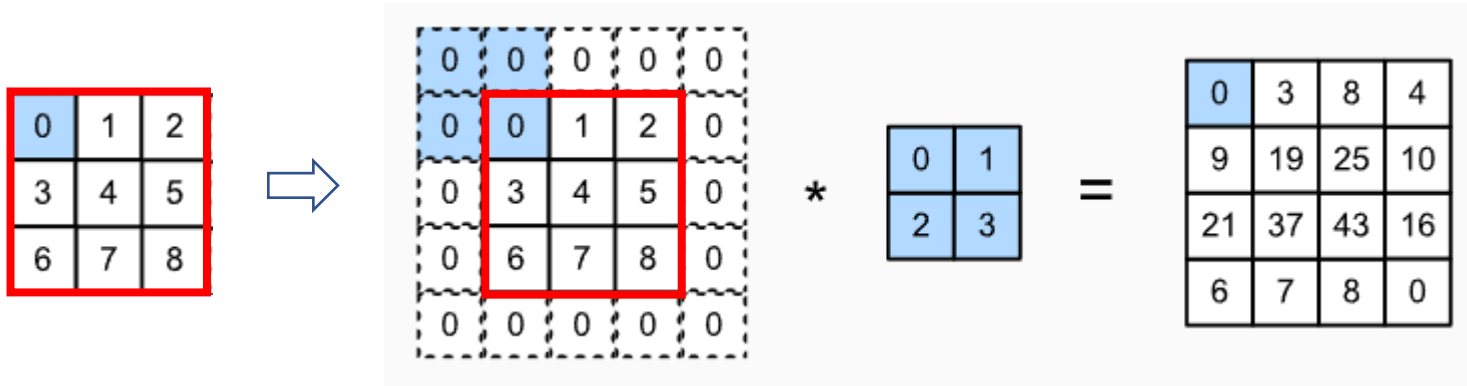
Convolution Layer – Practical Considerations

- Padding
  - When you convolve an image with a filter, we will loose pixels at the boundary of the image

- Solution → Padding
  - Add extra pixels (set to 0) of filler around the boundary of the input image
  - Typically, we use kernels with odd height and width, e.g., 1, 3, 5, 7



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Convolution Layer – Practical Considerations

- Typically,
  - The convolution window starts at the upper-left corner, and then we slide it over all locations both down and to the right
  - We slide the window one element at a time
  - For computational efficient (and to downsample)
    - Slide the window more than one elements, skipping the intermediate locations
  - The number of rows (and columns) skipped is called stride

- Hyper parameters
  - The kernel size
  - No. of output channels
  - Padding
  - The stride

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Illustration of Convolution with Stride

Given this 5x5 image, convolve with a 3x3 filter

1	2	2	3	1
1	1	3	3	2
3	2	1	2	3
2	3	2	3	2
2	1	2	3	1

1	2	1
1	2	1
1	2	1

[OFFICIAL]

Illustration of Convolution with Stride

Given this 5x5 image, convolve with a 3x3 filter **with Stride = 2**

1	2	1	3	1
1	2	1	3	2
1	2	1	2	3
2	3	2	3	2
2	1	2	3	1

Move the filter over the entire image from left to right, top to bottom

1x1 + 2x2 + 1x2 + 1x1 + 2x1 + 1x3 + 1x3 + 2x2 + 1x1

21	

1	2	2	3	1
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[OFFICIAL]

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2	1	2	3	1

Move the filter over the entire image from left to right, top to bottom **skipping 2 columns or rows respectively**

1x2 + 2x3 + 1x1 + 1x3 + 2x3 + 1x2 + 1x1 + 2x2 + 1x3

21	28

1	2	2	3	1
1	1	3	3	2
3	2	1	2	3
2	3	2	3	2
2	1	2	3	1

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Illustration of Convolution with Stride

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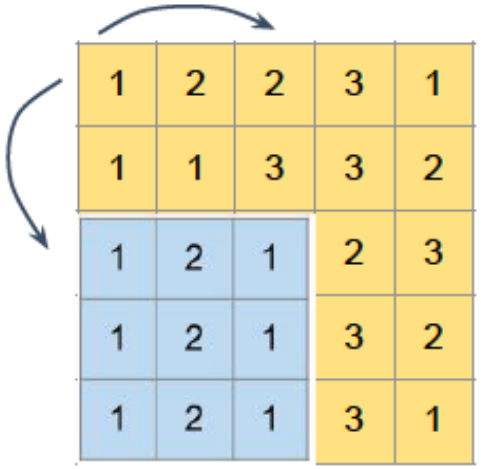

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1x3 + 2x2 + 1x1 + 1x2 + 2x3 + 1x2 + 1x2 + 2x1 + 1x2

21	28
24	

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[OFFICIAL]

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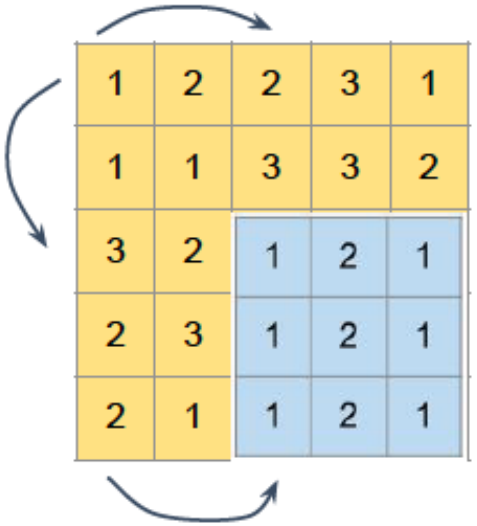

Move the filter over the entire image from left to right, top to bottom **skipping 2 columns or rows respectively**

1x1 + 2x2 + 1x3 + 1x2 + 2x3 + 1x2 + 1x2 + 2x3 + 1x1

21	28
24	27

1	2	2	3	1
1	1	3	3	2
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Pooling Layer

- Pooling is the operation of aggregating the values within a window around a pixel to produce one singe value, e.g.,
  - Max pooling
    - takes the maximum of all the values within a window around a pixel
  - Min pooling
    - takes the minimum of all the values within a window around a pixel
  - Average pooling
    - takes the minimum of all the values within a window around a pixel

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  - Min pooling
    - takes the minimum of all the values within a window around a pixel
  - Average pooling
    - takes the minimum of all the values within a window around a pixel
- It is important when stacking multiple layers
  - Reduces the size of the output of a layer before feeding it to the next layer
- Advantages
  - Reduce the size of the feature map
  - Although the size of the filters remain the same, the receptive field will increase with the depth of the network – thus subsequent layers will capture bigger context

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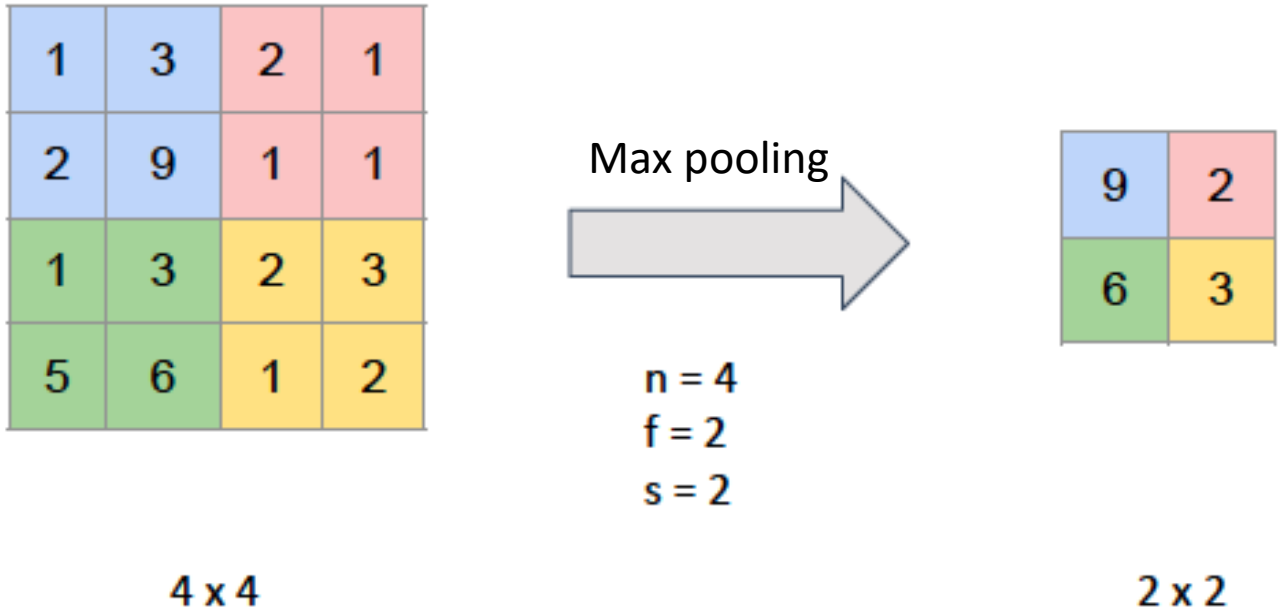
Illustration of Max Pooling



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In this Lecture

- Introduction
  - Linear regression, perceptron and MLP
  - Why we need CNNs
- Convolutional Neural Network (CNN)
  - Convolutions
  - Convolution layers
- Examples of CNNs
  - LeNet
- Summary

- Learning objectives
  - Understand CNN and the different components that compose it
  - Implement CNNs in Python, NumPy and PyTorch
- Additional readings
  - Chapter 7 of the textbook, available at: https://d2l.ai/

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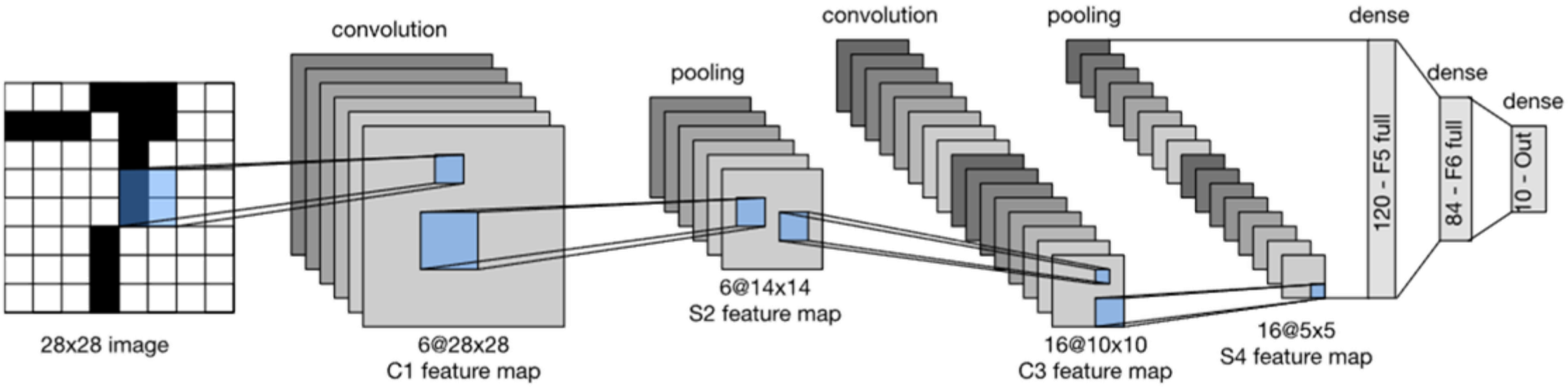
Convolutional Neural Networks: LeNet

- LeNet, the first published CNN (by Yann LeCun in 1998), used to recognize handwritten digits in images
  - Used to recognize digits for processing deposits in ATM machines (1990s)
  - Some ATM machines are still using this code!



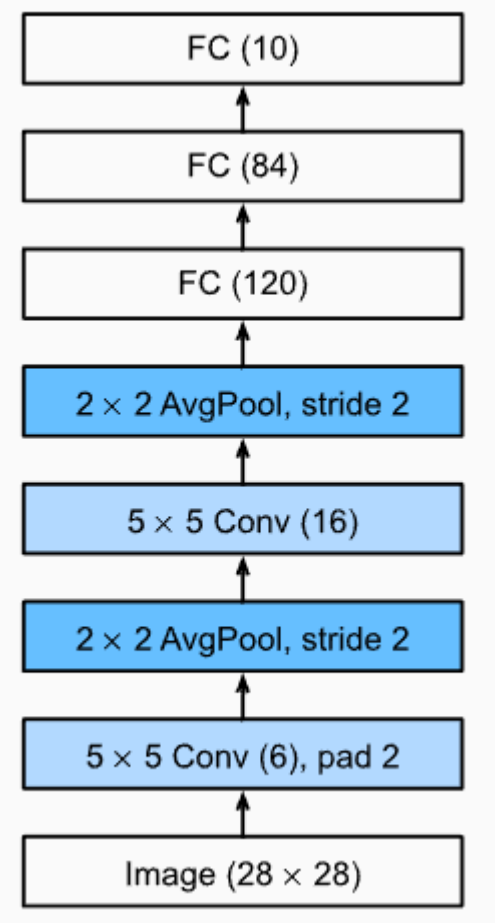
Convolutional encoder composed of 2 convolutional layers (5x5 kernel), each layer has sigmoid activation and is followed by 2x2 average pooling and stride 2

Two fully connected layers

https://d2l.ai/chapter_convolutional-neural-networks/padding-and-strides.html

Convolutional Neural Networks: LeNet

- LeNet, the first published CNN (by Yann LeCun in 1998), used to recognize handwritten digits in images
  - Also used to recognize digits for processing deposits in ATM machin ATM machines are still using this code!
- In the lab,
  - You will create the network and train it on the MNIST dataset



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https://d2l.ai/chapter_convolutional-neural-networks/padding-and-strides.html

Summary

- We derived the structure of convolutional layers
- In the lab
  - You will create and train LeNet
- Next week
  - Training (including data preparation), validation and testing
  - Tuning the hyper parameters
  - Modern Convolutional Neural Networks

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Questions

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